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1. Introduction

Industrial practice is an essential part of the educational process aimed at developing practical skills in system analysis, backend development, and AI integration within a real-world project environment. This report presents the results of the industrial practice completed as part of the Software Engineering educational program.

The practice was focused on participating in the development of a modern Learning Management System (LMS) called *Sana*, which is designed for corporate training and employee development. The platform integrates modules such as course management, training requests, certification tracking, and learning analytics, providing a complete ecosystem for employee development.

During the internship, special attention was given to the design and development of an AI assistant based on Large Language Models (LLM), integrated into the LMS platform. The assistant is intended to support users by providing guidance, answering questions, and simplifying interaction with the system.

1.1. Place of practice

The industrial practice was completed remotely in collaboration with **PetroSkills Pro LLP**, which is involved in the development of a startup project — *AI-LMS*, an intelligent learning management system for corporate training.

The project is developed in cooperation with the company *Nomad Speakers*, which specializes in organizing corporate training, business events, and implementing digital learning solutions in Kazakhstan.

The *Sana* platform represents a comprehensive LMS system designed to support corporate education processes, improve employee competencies, and provide scalable and flexible learning solutions using modern technologies.

1.2. Goals and objectives

The main goal of the industrial practice was to gain practical experience in software development, with a focus on backend systems, system architecture, and AI integration within a corporate LMS platform.

The key objectives of the practice included:

- studying the architecture and functionality of the Sana LMS platform
- analyzing AI-chat architectures based on Encore framework
- comparing modern LLM models (Gemini, GPT, Claude) for integration into the system
- designing an AI assistant adapted for LMS use cases

- developing system prompts for user-friendly AI interaction
- 🔶 participating in the development of backend API endpoints for AI communication
- integrating AI capabilities into the LMS environment
- improving skills in backend and frontend development
- working collaboratively within a development team

1.3. Expected results of industrial practice

As a result of the industrial practice, it was expected to develop both technical and professional competencies in software engineering, including system analysis, architecture design, and AI integration.

The expected outcomes included:

- development of a functional LMS platform prototype
- implementation of an AI assistant integrated into the system
- 🔶 development of an MVP AI chat prototype (in progress), capable of generating responses to user queries
- gaining experience in working with modern technologies such as Go (Encore), Next.js, and LLM-based systems
- understanding real-world software development processes and teamwork

The practice also aimed to enhance problem-solving skills, improve technical knowledge, and prepare for future professional activities in the field of software engineering.

2.2. Practical part: Development of AI Assistant for Sana LMS

During the industrial practice, a significant part of my work was focused on the research, design, and partial implementation of an AI assistant integrated into the Sana LMS platform.

At the initial stage, I participated in the MVP development in the role of a system analyst. My responsibilities included designing user flows, defining interface logic (notifications, attendance tracking, empty states), and reviewing frontend code for security aspects such as proper use of environment variables. Additionally, I contributed to documentation and prepared intermediate presentations to support the development process.

2.2.1. Research and analysis of AI solutions

Further work was focused on analyzing the possibility of integrating an AI assistant into the LMS platform. The main objective of the assistant is to support users by answering questions, explaining system functionality, and providing personalized recommendations for learning.

I analyzed the architecture of existing AI-based systems, including the *ai-chat* project built on the Encore framework. Special attention was given to understanding the interaction between components such as chat orchestration, LLM services, and provider layers.

Additionally, I conducted a comparative analysis of modern Large Language Models (LLMs), including Claude, GPT-4o, and Gemini. The evaluation was based on key criteria such as response quality, speed, cost, and the ability to handle business logic scenarios.

As a result of this analysis, Gemini was selected as the primary model for MVP implementation due to its balance between performance and cost efficiency.

2.2.2. Design of AI assistant architecture

Based on the research, I designed a simplified architecture of the AI assistant tailored to LMS requirements.

The architecture follows a request-response model:

Next.js → /assistant/chat → backend (Encore, Go) → Gemini → response

In this architecture:

- the frontend (Next.js) sends user queries to the backend
- the backend processes the request and enriches it with business context (such as user role, course data, and learning progress)
- the request is sent to the Gemini model
- the generated response is returned to the user

A separate assistant service was designed to handle AI-related logic, including prompt processing, context preparation, and communication with the LLM.

2.2.3. Prompt engineering and AI behavior design

An important part of my work was the development of a system prompt that defines the behavior, tone, and logic of the AI assistant.

The assistant was designed to act as a domain expert within the Sana platform, providing:

- clear and structured answers
- explanations of system functionality
- guidance for users based on their roles (employee, HR, admin)
- recommendations for courses and learning paths

Special attention was given to:

- ensuring user-friendly language
- avoiding technical jargon
- maintaining accuracy and consistency of responses
- structuring prompts to ensure stable and predictable AI behavior

Test scenarios were created to validate the behavior of the AI assistant, and the prompt was iteratively improved based on testing results.

2.2.4. Implementation of AI assistant (MVP)

At the implementation stage, I participated in the development of a prototype AI assistant.

This included:

- designing the backend endpoint for AI interaction (/assistant/chat)
- preparing system context, including project description and application structure
- 🔶 integrating the Gemini API into the backend service
- testing basic interaction scenarios between user and AI

The endpoint processes user queries, forwards them to the LLM, and returns structured responses to the frontend.

🔶 At the current stage, the MVP AI chat is in progress, and the backend integration with Gemini is being finalized.

2.2.5. Additional development tasks

In addition to the AI assistant, I developed a Python script to automatically parse the Next.js routing structure. This script extracts routes, functions, and imports from the project, which helps the AI assistant better understand the application structure and provide more relevant responses.

This approach improves the contextual awareness of the assistant and supports more accurate answers related to system navigation and functionality.

2.2.6. Technologies used

During the practice, the following technologies were used:

- Go (Encore) — backend development
- Next.js — frontend development
- Gemini API — integration of AI capabilities
- Python — auxiliary scripts for data processing
- Git / VS Code — development and collaboration tools

2.2.7. Challenges and solutions

During the development process, several challenges were encountered:

- Understanding AI architecture — initially, it was difficult to understand how LLM-based systems are structured. This was resolved through analysis of existing solutions such as ai-chat.
- Choosing the appropriate LLM model — different models were evaluated to find the optimal balance between performance and cost.
- Working with API integration — difficulties were encountered when configuring and testing API requests, which required additional debugging and experimentation.
- Designing effective prompts — it was necessary to structure prompts in a way that ensures consistent and accurate responses from the AI assistant.

These challenges contributed to gaining deeper practical knowledge and improving problem-solving skills.

3. Conclusion

The industrial practice provided valuable experience in applying theoretical knowledge to a real-world software engineering project. During the internship, I was involved in the development of a corporate Learning Management System (Sana) and contributed to both analytical and technical aspects of the project.

One of the key outcomes of the practice was the design and development of an AI assistant integrated into the LMS platform. I analyzed AI architectures, compared modern LLM models, designed system prompts, and contributed to the backend logic for AI interaction. This experience allowed me to understand how AI technologies can be effectively applied in real business systems.

In addition, I gained practical experience working with modern technologies such as Go (Encore), Next.js, and LLM-based APIs. I also strengthened my skills in system analysis, architecture design, and collaborative development within a team environment.

🔶 The development of the MVP AI assistant is currently in progress, and further work is focused on completing backend integration and improving interaction scenarios.

Overall, the practice significantly enhanced my understanding of building scalable systems and integrating AI into real-world applications. It also improved my ability to solve complex problems, work with new technologies, and adapt to project requirements.

The knowledge and experience gained during this practice will be valuable for my future professional development in software engineering, particularly in backend development and AI-driven systems.

4. Reflection

During the industrial practice, I gained valuable experience that significantly expanded my understanding of software engineering and real-world project development.

At the beginning of the internship, one of the main challenges for me was understanding the architecture of a complex system and how its components interact. Working on the Sana LMS project helped me develop system thinking and understand how frontend, backend, and AI components are integrated within a single application.

A particularly important part of my experience was working with AI technologies. Initially, I had limited practical experience with Large Language Models, but during the practice I analyzed different models, compared their capabilities, and selected the most suitable one for the project. I also designed system prompts and learned how prompt structure directly affects the quality and consistency of AI responses.

Another valuable aspect was working with backend-related tasks and external APIs. I was involved in designing the interaction flow of the AI assistant and preparing the backend endpoint for communication with the LLM. Although this process included challenges related to API integration and system architecture, I was able to overcome them through testing, debugging, and continuous learning.

I also improved my ability to work in a team environment, communicate ideas clearly, and contribute to shared project decisions. Participation in discussions, documentation, and collaborative tasks gave me a better understanding of real-world development processes.

🔶 At this stage, the MVP of the AI assistant is still in progress, but I already have a clear understanding of its architecture, interaction flow, and further development steps.

Overall, this practice helped me become more confident in working with complex systems and strengthened my practical skills in AI integration and backend development. It also helped me identify key areas for further growth, particularly in designing scalable systems and working with AI-driven solutions.

This experience confirmed my interest in software engineering and motivated me to continue developing in the field of AI-based applications.

5. References

1. Encore Documentation. Available at: https://encore.dev/docs

2. Next.js Documentation. Available at: https://nextjs.org/docs

3. Google Gemini API Documentation. Available at: https://ai.google.dev

4. Google Cloud Vertex AI Documentation. Available at: https://cloud.google.com/vertex-ai

5. OpenAI. Introduction to Large Language Models. Available at: https://platform.openai.com/docs

6. Anthropic Claude Documentation. Available at: https://docs.anthropic.com

7. Encore AI Chat Example Repository. Available at: https://github.com/encoredev/examples

6. Practice task

During the industrial practice, I was assigned tasks related to the development of a Learning Management System (Sana) with a focus on system analysis and AI integration.

My main responsibilities included:

- analyzing the architecture and functionality of the Sana LMS platform
- participating in MVP development as a system analyst
- designing user flows and interface logic (notifications, attendance tracking, empty states)
- reviewing frontend code for basic security practices (environment variables usage)
- researching AI assistant integration into LMS systems
- analyzing AI-chat architecture based on the Encore framework
- comparing modern Large Language Models (Gemini, GPT, Claude)
- designing the architecture of an AI assistant for LMS
- developing system prompts for AI interaction
- 🔶 participating in backend development of the AI assistant (API endpoint /assistant/chat)
- 🔶 preparing integration of the Gemini API
- testing AI interaction scenarios and improving prompt quality

These tasks allowed me to gain both analytical and practical experience in software development and AI system design.